


```
%Semester one
%Course Units Matrix
Course_units=["Engineering Mathematics III","Hydrology I","Soil Mechanics","Mechanics of Materials","Computer Programming","Engineering Ge
%Calculating the number of course Units
CUs_Number=length(Course_units);
%Entering the Marks into the Matrix
Marks=zeros(1,CUs_Number);
%Entering of the marks into the Matrix
for index=1:CUs_Number
    Marks(index)=input(['Enter your Marks for',Course_units{index},':']);
end%Display the Marks

disp('Your marks');
```

Your marks

```
disp(Marks);
```

75 64 76 64 59 67

```
%Grading system
Grade=zeros(1,CUs_Number);
GradingRange=[80,75,70,65,60,55,50];
GradingSystem=[5,4.5,4,3.5,3,2.5,2,0];
for index=1:CUs_Number
    Current_Mark=Marks(index);
    Grade(index)=GradingSystem(end);
    for g=1:length(GradingRange)
        if Current_Mark>=GradingRange(g)
            Grade(index)=GradingSystem(g);
            break;
        end
    end
end
%Displaying Results
disp('your Grades:');
```

your Grades:

```
disp(Grade);
```

4.5000 3.0000 4.5000 3.0000 2.5000 3.5000

```
%Update the marks array
%Credit Units
Credit_Units=[4 4 4 4 4 4];
%Calculate the total CUs
Total_CUs=sum(Credit_Units,"all","double");
GPA=(Grade*Credit_Units')/Total_CUs;
disp("your GPA is: ");
```

your GPA is: '

```
disp(GPA)
```

3.5000

```
%semester two
%Course Units Matrix
Course_units=["Engineering Mathematics II","Hydrology","Engineering Mechanics","Computer Applications","Circuit Theory","Engineering Drawi
%Calculating the number of course Units
CUs_Number=length(Course_units);
%Entering the Marks into the Matrix
Marks=zeros(1,CUs_Number);
%Entering of the marks into the Matrix
for index=1:CUs_Number
    Marks(index)=input(['Enter your Marks for',Course_units{index},':']);
end%Display the Marks

disp('Your marks');
```

Your marks

```
disp(Marks);
```

78 75 62 68 54 61 51

```
%Grading system
Grade=zeros(1,CUs_Number);
GradingRange=[80,75,70,65,60,55,50];
GradingSystem=[5,4.5,4,3.5,3,2.5,2,0];
for index=1:CUs_Number
    Current Mark=Marks(index);
```

```
Grade(index)=GradingSystem(end);
for g=1:length(GradingRange)
    if Current_Mark>=GradingRange(g)
        Grade(index)=GradingSystem(g);
        break;
    end
end
end
%Displaying Results
disp('your Grades:');
```

your Grades:

```
disp(Grade);
```

4.5000 4.5000 3.0000 3.5000 2.0000 3.0000 2.0000

```
%Update the marks array
%Credit Units
Credit_Units=[4 3 4 3 4 4 4];
%Calculate the total CUs
Total_CUs=sum(Credit_Units,"all","double");
GPA1=(Grade*Credit_Units')/Total_CUs;
disp("your GPA1 is: ");
```

your GPA1 is: '

```
disp(GPA1)
```

3.1538

```
%Calculating CGPA
CGPA=(GPA+GPA1)/2
```

CGPA = 3.3269

```
disp("your CGPA is:")
```

your CGPA is:'

```
disp(CGPA)
```

3.3269